Nilakshi Mishra

AI/ML Engineer

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SUMMARY

Building AI-powered applications with LLMs, RAG, AI Agents, FastAPI and AWS.

EXPERIENCE

Software AI/ML Engineer - Ommify Technologies Pvt. Ltd.

Jan 2025 - Present

Designing and shipping AI-powered applications - LLM products, retrieval pipelines, and FastAPI services on AWS.

- Building LLM applications with RAG, agents, and tool calling.
- Developing FastAPI backends that connect models to real product workflows.
- Deploying and operating services on AWS with an eye for reliability.

AI Intern - CETPA Infotech Pvt. Ltd.

Sep 2024 - Jan 2025

Applied machine learning and Python to practical problems, building a foundation in model-driven features and AI application engineering.

- Worked on applied AI/ML tasks and Python-based development.
- Explored how models move from notebooks into usable application flows.

PROJECTS

AI Campaign Chatbot

A campaign assistant that answers from retrieved knowledge instead of guessing. FastAPI serves the conversation layer; OpenAI handles generation; FAISS keeps campaign context close to the query.

Stack: Python, FastAPI, OpenAI, RAG, FAISS

Event-Driven Notification Service

A notification backend that reacts to system events, persists delivery state in PostgreSQL, and uses Redis to coordinate fast, reliable fan-out.

Stack: Python, FastAPI, PostgreSQL, Redis

Enterprise Scraper Bot Factory

A distributed scraping platform: Celery workers pull jobs from Redis, results are indexed in OpenSearch, and artifacts land in MinIO for later use.

Stack: Python, Celery, Redis, OpenSearch, MinIO

SEO Optimization Tool

An SEO analysis service that uses LLMs for recommendations, FastAPI for the API layer, and PostgreSQL to store site and keyword data.

Stack: Python, FastAPI, PostgreSQL, AI/LLM

SKILLS

AI & Systems: FastAPI, LLM, RAG, LangChain, LangGraph, AI Agents, Tool Calling, MCP

Data & Retrieval: PostgreSQL, MongoDB, Redis, MySQL, Pinecone, FAISS

Languages: Python, JavaScript, C

Cloud & Tools: AWS, Docker, Git

Frontend: React.js

RESEARCH

Triple Band Miniaturized E-Shaped Multistrip Monopole Antenna for Multiband Wireless System

ICVMWT 2024, Springer

A compact E-shaped multistrip monopole designed for triple-band wireless operation, studied for Bluetooth, WiMAX, and mobile communication.

EDUCATION

B.Tech Electronics & Communication Engineering

Dr. Ram Manohar Lohia Awadh University | 2020 - 2024